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(base) alirezahamoudzadeh@Alirezas-MacBook-Air Demo % export OPENROUTER_API_KEY=sk-or-v1-6[REDACTED]
export AWS_ACCESS_KEY_ID=Q[REDACTED]
export AWS_SECRET_ACCESS_KEY=3[REDACTED]

cd /Users/alirezahamoudzadeh/Downloads/fire/MCP server + LLM Demo
python demo_client.py
cd: too many arguments

[model called tool] analyze_burn_severity_tool({'min_lon': 14.425, 'min_lat': 40.792, 'max_lon': 14.51, 'max_lat': 40.85, 'pre_start': '2025-05-09', 'pre_end': '2025-08-06', 'post_start': '2025-08-19', 'post_end': '2025-11-16'})
/opt/anaconda3/lib/python3.12/site-packages/rasterio/warp.py:387: NotGeoreferencedWarning: Dataset has no geotransform, gcps, or rpcs. The identity matrix will be returned.
  dest = _reproject(
[09/24/26 23:13:44] INFO Created 540 records raw.py:723

=== Grounded answer ===
Based on the analysis of satellite imagery for the wildfire near Terzigno in Vesuvius National Park, here are the key findings:

## Total Burned Area
**668.38 hectares** burned in total within the specified bounding box during the fire period.

## Burn Severity Distribution
- **Low severity:** 370.55 hectares (416 patches)
- **Moderate-low severity:** 264.9 hectares (93 patches)
- **Moderate-high severity:** 32.93 hectares (31 patches)

## Priority Areas for Erosion Control
**27.51 hectares** should be prioritized for erosion-control work before the rainy season, concentrated in **12 critical patches**. These high-priority areas represent the most vulnerable locations where soil stabilization measures are most urgently needed.

The relatively small size of the priority areas (27.51 ha) compared to the total burned area suggests that while the fire affected a large region, the most erosion-prone zones are concentrated in specific areas, likely steeper slopes or locations with particular soil conditions that make them more susceptible to runoff and debris flow during heavy rainfall.

**Recommendations:** Focus erosion-control efforts on these 12 priority patches, implementing measures such as hydroseeding, straw mulching, sediment barriers, and contour felling to stabilize soils before the rainy season begins.
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